Preprint

SUPPLEMENTARY MATERIAL The Locked Bimetric Potential and the Constitutive Law

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Abstract

Papers VIII and XII lock the five Hassan–Rosen (HR) coefficients to $\beta = (1, 0, 16, 0, 1)$ from the stellated-octahedron geometry, and Paper XII adds that the locked theory and the SST are two projections of one geometry, so the bimetric theory should give $\mu(x) = x/(x+1)$. We separate what holds from what does not. *The geometric lock largely holds.* Its exchange symmetry is exact: the $g \leftrightarrow f$ symmetry holds if and only if β is palindromic. Its odd coefficients vanish exactly under total inversion $S \to -S$, not spatial parity. And the partition it defines carries μ: the two volume terms, and every term weighted by its number of TetB legs, split in a ratio whose odds Ω are inverted by the metric exchange, so the TetB share is exactly $\mu(\Omega) = \Omega/(1+\Omega)$. The stellated octahedron ($S = P$) sits at the pole $\Omega = -1$, which is why the potential vanishes there. *The bridge to the SST does not hold.* μ needs a second ingredient: odds equal to the acceleration, $\Omega = x = g/a_0$. HR's potential has no derivatives, so in any weak field its odds stay at 1. In the standard reduction the response depends on the enclosed mass only through M/r^3, on every branch and for every β, and no function of M/r^3 equals a function of $x \propto M/r^2$. By Letter 2's Lemma, HR cannot even see the two roots in metric form ($g^{-1}f = I$). A frame deformed by hand along the time leg's acceleration supplies $\Omega = x$ and reproduces μ, but it is not an HR metric. *The result in two steps:* (1) exchange symmetry of the partition gives $\mu(\Omega)$, which HR contains; (2) $\Omega = x$, which HR cannot supply and the SST supplies through the time leg's acceleration.

Keywords: Hassan–Rosen, bimetric gravity, Vainshtein, Galileon, exchange symmetry, constitutive law, MOND, proof

Setting

The Hassan–Rosen (HR) interaction is $m^2\sqrt{-g}\,V(S)$, $V(S) = \sum_{n=0}^{4} \beta_n e_n(S)$, with $S = \sqrt{g^{-1}f}$ and e_n the elementary symmetric polynomials of its eigenvalues [1]. The lock is $\beta = (1, 0, 16, 0, 1)$ [2, 3]. The SST's scalar sector is $-(a_0^2/8\pi G)F(y)$, $F(y) = y - 2\sqrt{y} + 2\ln(1+\sqrt{y})$, $\mu(x) = F'(x^2) = x/(x+1)$ [7, 8]. The question, listed as open in Papers VIII and IX, is whether the first reduces to the second. Each step is checked by the accompanying script, 63 checks, exact algebra [12], including the decoupling-limit derivation itself, done independently of [11].

Part A: the standard reduction

Theorem 1 (the response depends on density). *In the decoupling limit, with Stückelberg field π and $\Lambda_3^3 = m^2 M_{\rm Pl}$, the h–π mixing has coefficients $d_0 = \beta_0 + 3\beta_1 + 3\beta_2 + \beta_3$, $d_1 = \beta_1 + 2\beta_2 + \beta_3$, $d_2 = \beta_2 + \beta_3$, $d_3 = \beta_3$; for the lock, $(49, 32, 16, 0)$. Around a static mass, with $d_3 = 0$, $\lambda = \pi'/r = -sq/(2Ma_1)$ and $Q = a_2 q/(Ma_1^2)$, the π-equation is*

$$Q^2 s^3 + 6Qs^2 + 6s - 2 = 0, \qquad g_{\rm tot}/g_N = 1 + s, \tag{1}$$

with $q = M_{\rm enc}/4\pi r^3$. The radius enters only through q.

Proof. The identity $\sum_n (-1)^n \beta_n Y_n(1 - \Pi) = \sum_k d_k Y_k(\Pi)$, with $Y_n(X) = \sum_k (-1)^k e_k(X) X^{n-k}$, is checked for general β on an exact random symmetric Π ($Y_4 = 0$ by Cayley–Hamilton). From $-\frac{1}{2}h\mathcal{E}h + h \cdot \sum_k a_k Y_k(\Pi) + h_{00}\rho/2M$ with $h = \mathrm{diag}(A, B, B, B)(r)$ and $\pi(r)$, all three Euler–Lagrange equations are r^{-2} times the derivative of a current (checked), so they integrate exactly. Regularity at the origin gives the π-equation $\frac{3}{2}a_1^2\lambda - (3a_1a_2 + a_3 q/4M)\lambda^2 + (a_2^2 + \frac{2}{3}a_1a_3)\lambda^3 - \frac{1}{6}a_3^2\lambda^5 + a_1 q/4M = 0$, which contains no r, and $g_{\rm tot}/g_N = 1 - 2Ma_1\lambda/q + \frac{2}{3}Ma_3\lambda^3/q$ (checked). Setting $a_3 = 0$ and substituting gives (1). □

$d_1 = 32$ is the Fierz–Pauli combination and $d_3 = 0$ removes the quintic mixing; $d_2 = 16$ keeps the cubic Galileon term (Paper XII states that both vanish).

Theorem 2 (the branches). *Every real root of (1) is $s = 2/D(v)$, $Q = vD(v)/2$, with $D = v^2 + 6v + 6$.*

1. *The branch that matches the linear solution ($v \to 0$: $s = 1/3$, $g_{\rm tot}/g_N = 4/3$, vDVZ) is screened for $Q > 0$ ($s \to 0$). For $Q < 0$ it turns at $Q_* = 2 - 2\sqrt{2}$, where $s = 1/\sqrt{2}$, and does not continue past it. On it, $g_{\rm tot}/g_N \le 1 + 1/\sqrt{2} < 1.71$.*
2. *Where $D < 0$, $s \le -2/3$, so $g_{\rm tot}/g_N \le 1/3$.*

3. *As $D \to 0^+$ ($v \to -3 \pm \sqrt{3}$), $Q \to 0$ and $s \simeq v/Q \to \infty$. There $g_{\rm tot} \simeq g_N v/Q \propto (M/r^2)(r^3/M) = r$, independent of the mass.*

Proof. Substituting the parametrisation solves (1) identically; the turning points are $v = -2 \pm \sqrt{2}$, $\min D = -3$, and the roots of D are $-3 \pm \sqrt{3}$, where $Q = 0$ (all checked). □

The unbounded roots of (1) do not match the linear solution at large r, and their force grows like r whatever the mass: a self-accelerating-type branch, not the deep-MOND form $\sqrt{GMa_0}/r$. So no branch gives flat rotation curves. Theorem 3 shows this for every branch at once.

Theorem 3 (no change of variable). *No function of M/r^3 equals a function of $x = GM/(a_0 r^2)$ for all M and r, unless both are constant. Since every root of the π-equation is a function of q alone (Theorem 1), the standard response cannot be $1/\mu(x)$ on any branch, for any β.*

Proof. Along $M = Lr^3$ the density is fixed while $x = GLr/a_0$ varies, and $\mathrm{d}[1/\mu(x)]/\mathrm{d}r \neq 0$ (checked). Equivalently, the Vainshtein radius scales as $M^{1/3}$ and the MOND radius $\sqrt{GM/a_0}$ as $M^{1/2}$. □

With $m \sim H_0$, galaxies also lie well inside the Vainshtein radius (∼260 kpc at $10^{10} M_\odot$), where $\mu_{\rm eff} \to 1$ [11]. Dimensionally, the Galileon variable is $\Phi/(mr)^2$ and the MOND variable $\Phi/(mr)$: both small quantities Φ/c^2 and rH_0/c are $\sim 10^{-6}$ at galactic radii, which is why $a_0 \sim cH_0$ is natural, but the potential's strongest interactions select the wrong power.

Theorem 4 (the parity configuration and the polynomial). *At $S = P = \mathrm{diag}(1, -1, -1, -1)$, $e_n(P) = (1, -2, 0, 2, -1)$ and $V(P) = \beta_0 - \beta_4 = 0$. On $S = \mathrm{diag}(1, s, s, s)$ the locked potential is $1 + 48s + 48s^2 + s^3$ (98, 1, 0 at $s = +1, 0, -1$), not the coupling polynomial $1 + 4s + 16s^2$ (21, 17, 13).*

Proof. Direct evaluation (checked). □

So GFD's two roots (one metric, $S = P$) and HR's two metrics are different objects, and the potential's values are not the polynomial's.

Theorem 5 (Hassan–Rosen is root-blind). *TetA $= e$ and TetB $= Pe$ give the same metric, $g = e^{\mathsf{T}} \eta e = (Pe)^{\mathsf{T}} \eta (Pe)$, while $\det e$ changes sign. Hence, with f built from TetB, $g^{-1} f = I$, and the HR potential takes the same value on either root. Only the vielbein form $e^{-1}(Pe) = P$ sees the second root, and there $V(P) = 0$.*

Proof. $P^{\mathsf{T}} \eta P = \eta$; checked for a general static tetrad. □

The two roots are the two square-root branches of $g^{-1} f = I$: P satisfies $P^2 = I$ but is not the principal root (its spatial eigenvalues are -1), and $e^{\mathsf{T}} \eta (Pe)$ is symmetric, so the vielbein form with $S = P$ meets the symmetric-vielbein condition (checked). The metric form, defined with the principal root, sees only $S = I$.

This is Letter 2's Lemma [4]: objects built from the metric (the metric, a torsion-free connection) cannot tell TetA from TetB, and only the torsion sector can. The HR interaction is built from the metrics and contains no torsion, so it cannot carry the TetA/TetB partition. That is the reason for Theorems 1–4.

Part B: a deformed frame (diagnostic)

Theorem 6 (exchange closure on a deformed frame). *On $S = \mathrm{diag}(1, 1, 1, w)$, for every β,*

$$V = A + Bw,$$
$$A = \beta_0 + 3\beta_1 + 3\beta_2 + \beta_3,$$
$$B = \beta_1 + 3\beta_2 + 3\beta_3 + \beta_4.$$

The deformed leg's share $s(w) = Bw/(A + Bw)$ satisfies $s(1/w) = 1 - s(w)$ if and only if $A = B$, that is

$$\beta_0 + 2\beta_1 = \beta_4 + 2\beta_3, \tag{2}$$

and then $s(w) = w/(1 + w)$ exactly.

Proof. $s(1/w) - [1 - s(w)] = -w(A - B)(A + B)/[(A + Bw)(Aw + B)]$ (checked), which vanishes iff $A = B$ since $A + B > 0$. □

The share is not an arbitrary choice: $Bw/(A + Bw) = w\,\mathrm{d}\ln V/\mathrm{d}w$, the elasticity of V on this family (checked).

Corollary 1 (the lock and β_2). *HR's $g \leftrightarrow f$ symmetry $\det S\, V(S^{-1}) = V(S)$ holds iff β is palindromic (the difference is $\sum_n (\beta_{4-n} - \beta_n) e_n(S)$). Palindromic β satisfies (2), and then $V = (\beta_0 + 4\beta_1 + 3\beta_2)(1 + w)$. The locked set gives $V = 49(1 + w)$, and β_2 cancels from s. On this family $V/V(I) = \det[(I + S)/2]$, the volume of the midpoint frame.*

Proof. All identities checked exactly; $(1, 0, 16, 0, 2)$ shows that (2) is a real condition. □

The share's form follows from the $g \leftrightarrow f$ symmetry, not from the value 16.

Remark (not the two chambers). The two shares $w/(1 + w)$ and $1/(1 + w)$ sum to 1, have odds w, and swap under $w \to 1/w$: the algebra of the partition. But they are the deformed leg and the three undeformed legs of

one frame, not TetA and TetB. On the actual roots the locked potential is $V(I) = 98$ and $V(P) = 0$, and on the path between them, $\mathrm{diag}(1, s, s, s)$, it is $(1 + s)(s^2 + 47s + 1)$, with no exchange symmetry between the ends (checked). So this is not the two-chamber mechanism of [10, 4].

Theorem 7 (the time leg, and what the potential does not supply). *(i) With $w = |a|/a_0$, a the acceleration of the tetrad's time leg, a static field has $c^2|a| = |\nabla\Phi|$ exactly [9], so $w = x$ and $s = \mu(x)$. (ii) The auxiliary energy*

$$E(u; y) = (1 - e^{-u})\, y - 2(\sinh u - u),$$

with $u = \ln(V/V_0)$, is stationary at $V/V_0 = 1 + \sqrt{y}$, where its value is $F(y)$ and its slope $\partial E/\partial y$ is μ. (iii) Its u-dependence contains no β.

Proof. All checked exactly. □

So the potential supplies the variable $V/V_0 = 1 + x$; the energy F, and with it the dynamics, comes from the SST's own scalar term.

Part C: what the lock establishes

Write Ω for the odds between a TetB share and a TetA share, so that the TetB share is $\Omega/(1 + \Omega)$. Letter 2's supplement [4] derives μ from two ingredients: an exchange that inverts Ω, and odds of degree one in the acceleration, $\Omega = x$. This part asks which of the two the locked potential supplies.

Theorem 8 (the partition carries μ). *For palindromic β:*

1. *the volume terms $\beta_0 e_0$ (four TetA legs) and $\beta_4 e_4 = \beta_0 \det S$ (four TetB legs) stand in the ratio $1 : \det S$, so with $\Omega = \det S$ the TetB share of the pair is $\Omega/(1 + \Omega)$;*

2. *weighting every term by its number of TetB legs gives the share $\sum_n n\beta_n e_n/(4V)$, which in its own odds is exactly $\Omega/(1 + \Omega)$, for every S and every β_2;*

3. *in both cases the metric exchange $S \to S^{-1}$ inverts Ω and swaps the shares, and total inversion $S \to -S$ leaves them unchanged (when $\beta_1 = \beta_3 = 0$).*

Proof. All identities checked for general eigenvalues (checks E1–E9). □

Theorem 9 (which operation exchanges the chambers). *(i) The odd coefficients vanish iff $V(-S) = V(S)$, since $V(-S) - V(S) = -2(\beta_1 e_1 + \beta_3 e_3)$: the lock is symmetric under total inversion, $V(-I) = V(I) = 98$, not under spatial parity, $V(P) = 0$. (ii) Spatial parity of one tetrad negates the odds, $\Omega \to -\Omega$; only the metric exchange inverts them. (iii) The stellated octahedron $S = P$ has $\Omega = \det P = -1$, the pole of $\Omega/(1 + \Omega)$: its two volume terms cancel and $e_2(P) = 0$, which is why $V(P) = 0$. P is its own inverse, so it is a fixed point of the exchange. The total-inversion image $-I$ has $\Omega = 1$: equal shares, the ring $x = 1$.*

Proof. Checked (E2–E5, E13, E14). □

So the exchange that Letter 2's derivation needs acts, in HR's algebra, as the metric exchange, not as spatial parity; the same distinction appears in the null-rotation proof [6], where TetB's count is reproduced by total inversion.

Theorem 10 (the odds are not the acceleration). *For two static weak-field metrics, $S = \mathrm{diag}(n, 1, 1, 1)$ with $n = \hat{N}/N$. The volume odds are n; the locked leg odds are $(25n + 24)/(24n + 25)$, confined to $[24/25, 25/24]$. In the weak field $n = \exp[(\hat{\phi} - \phi)/c^2] \to 1$, so the odds stay at 1 and the shares at $\frac{1}{2}$: μ is reached only at $x = 1$. Reaching every x requires $\Omega = n = g/a_0$, a second lapse built from the gradient of the first, which is the deformed frame of Part B.*

Proof. Checked (E10–E12). □

Corollary 2 (β_2). *The Hessian of e_2 over the 16 entries of S has rank 16 and spectrum $+3^1, +1^6, -1^9$, as Paper XII states (E15). But β_2 drops out of $\mu(\Omega)$ (Theorem 8), and Paper XII itself takes $\beta_2 = 16$ as an axiom; its effects are on the background, the graviton mass and the Higuchi margin (Part D of the script).*

Conclusion and scope

- *What holds.* The lock's exchange symmetry is exact (Corollary 1), its odd terms vanish by total inversion (Theorem 9), and its partition carries $\mu(\Omega) = \Omega/(1 + \Omega)$ in its own odds, with the metric exchange inverting them (Theorem 8). The stellated octahedron is the pole $\Omega = -1$. The lock also fixes bimetric physics: the background, the Fierz–Pauli mass, the Higuchi margin and the screening cubic.

- *What does not.* The locked HR theory does not give $\mu(x)$: its odds are not the acceleration (Theorem 10); in the standard reduction the response depends on M/r^3 on every branch, for every β (Theorems 1–3); and HR is root-blind (Theorem 5). Papers VIII, IX and XII overstate the equivalence.

- *The result in two steps.* (1) Exchange symmetry of the partition gives $\mu(\Omega)$; HR contains this. (2) $\Omega = x$; HR cannot supply it, and the SST supplies it through the time leg's acceleration [9]. Part B's deformed frame works only because it builds in step (2) by hand.
- *Open.* An action that supplies $\Omega = x$ from the two chambers themselves. Letter 2 places the partition in the torsion sector, where the time leg's acceleration is the torsion component $T^0{}_{i0}$; that is the subject of [5]. Also not done here: the full bimetric (dynamical f) decoupling limit.

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